

LEVEL 9 — Aggregation Functions

Easy line-by-line explanations with examples and outputs

First line

```
import numpy as np
```

1. np.sum()

Adds all values.

```
arr=np.array([10,20,30,40])  
print(np.sum(arr))
```

Output: 100

2. np.prod()

Multiplies all values.

```
arr=np.array([2,3,4])  
print(np.prod(arr))
```

Output: 24

3. np.cumsum()

Returns cumulative sums.

```
arr=np.array([10,20,30,40])  
print(np.cumsum(arr))
```

Output: [10 30 60 100]

4. np.cumprod()

Returns cumulative products.

```
arr=np.array([2,3,4])  
print(np.cumprod(arr))
```

Output: [2 6 24]

5. np.min()

Finds the smallest value.

```
arr=np.array([10,5,30,2,20])  
print(np.min(arr))
```

Output: 2

6. np.max()

Finds the largest value.

```
arr=np.array([10,5,30,2,20])  
print(np.max(arr))
```

Output: 30

7. np.mean()

Calculates average.

```
arr=np.array([10,20,30,40])  
print(np.mean(arr))
```

Calculation: $(10+20+30+40)/4 = 25$

Output: 25.0

8. np.median()

Finds the middle value after sorting.

```
arr=np.array([10,40,20,30,50])
```

```
print(np.median(arr))
```

Sorted: [10,20,30,40,50]

Output: 30.0

9. np.std()

Measures spread around the mean.

```
arr=np.array([10,20,30,40])
```

```
print(np.std(arr))
```

Output: approximately 11.1803

10. np.var()

Calculates variance (standard deviation squared).

```
arr=np.array([10,20,30,40])
```

```
print(np.var(arr))
```

Output: 125.0

11. np.ptp()

Returns range = maximum minus minimum.

```
arr=np.array([10,20,30,40])
```

```
print(np.ptp(arr))
```

40-10 = 30

Output: 30

12. axis=0

For a 2D array, axis=0 works down rows and gives one result per column.

```
arr=np.array([[10,20,30],[40,50,60]])
```

```
print(np.sum(arr,axis=0))
```

Output: [50 70 90]

10+40=50, 20+50=70, 30+60=90.

13. axis=1

For a 2D array, axis=1 works across columns and gives one result per row.

```
arr=np.array([[10,20,30],[40,50,60]])
```

```
print(np.sum(arr,axis=1))
```

Output: [60 150]

14. keepdims=True

Keeps the reduced dimension.

```
arr=np.array([[10,20,30],[40,50,60]])
```

```
result=np.sum(arr,axis=1,keepdims=True)
```

```
print(result)
```

Output:

```
[[ 60]]
```

```
[150]]
```

Without keepdims: [60 150].

15. axis=0 + keepdims=True

Keeps a 2D result while calculating column-wise.

```
arr=np.array([[1,2,3],[4,5,6]])
```

```
print(np.mean(arr,axis=0,keepdims=True))
```

Output:

```
[[2.5 3.5 4.5]]
```

Quick Revision

sum → total | prod → product | cumsum → cumulative total | cumprod → cumulative product

min → smallest | max → largest | mean → average | median → middle

std → standard deviation | var → variance | ptp → max-min

axis=0 → column-wise result | axis=1 → row-wise result | keepdims=True → keeps reduced dimension